

Geometrically, this is a plane in $\mathbb{R}^3$. $\therefore \text{Span}(u, v) = \text{a plane in } \mathbb{R}^3$.

Let, $w = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$ be in $\text{Span}(u, v)$. By defn, $\exists p, q \in \mathbb{R}$ s.t. $\boxed{pu + qv = w}$

$$\therefore p \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + q \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} \quad \therefore \text{From our soln. of previous portion, } p = \frac{6a_1 - b_1}{17}, q = \frac{3b_1 - a_1}{17} \text{ and } c_1 = 0$$

$$\therefore 17p = 6a_1 - b_1 \text{ and } 17q = 3b_1 - a_1 \quad \therefore 51p + 17q = 17a_1 \Rightarrow \boxed{a_1 = 3p + q}$$

$$\text{and } c_1 = 0. \quad 51p = 18a_1 - 3b_1 \quad \therefore 51p + 17q = 17a_1 \Rightarrow \boxed{b_1 = p + 6q}$$

$$17q = 3b_1 - (3p + q) \Rightarrow 17q = 3b_1 - 3p - q \Rightarrow \boxed{b_1 = p + 6q}$$

$\Rightarrow$ From Theorem 7, we can conclude that $\{u, v \text{ and } w\}$ are linearly dependent.

Let, $\{u, v, w\}$ be linearly dependent. $\therefore$ There exists weights c_1, c_2 and c_3 s.t. all are not zero together where $c_1 u + c_2 v + c_3 w = 0$

$$c_1 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

By defn. of Matrix equation $Ax = b$, we have $\begin{bmatrix} 3 & 1 & a_1 \\ 1 & 6 & b_1 \\ 0 & 0 & c_1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 From Theorem 3, the matrix equation $Ax = b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{bmatrix} 3 & 1 & a_1 & 0 \\ 1 & 6 & b_1 & 0 \\ 0 & 0 & c_1 & 0 \end{bmatrix} \xrightarrow[\text{(From prev part)}]{\text{Reduction algorithm}} \begin{bmatrix} 1 & 0 & (6a_1 - b_1)/17 & 0 \\ 0 & 1 & (3b_1 - a_1)/17 & 0 \\ 0 & 0 & c_1 & 0 \end{bmatrix}$$

$$c_1 + c_3 \left(\frac{6a_1 - b_1}{17} \right) = 0 \quad (2) \quad c_2 + c_3 \left(\frac{3b_1 - a_1}{17} \right) = 0 \quad (3) \quad c_1 c_3 = 0$$

$$\text{If } c_3 = 0, \text{ then } c_2 + 0 \left(\frac{3b_1 - a_1}{17} \right) = 0 \Rightarrow c_2 = 0$$

$c_1 + 0 \left(\frac{6a_1 - b_1}{17} \right) = 0 \Rightarrow c_1 = 0$. This is a contradiction to our initial assumption

$\therefore c_3 \neq 0$. From (3), $c_1 = 0$.